

PRIVATE SHIELD: Review of Privacy-Preserving Federated Learning for IoT Intrusion Detection

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Abstract—PRIVATE SHIELD examines federated intrusion detection for heterogeneous Internet of Things (IoT) networks. In CSE 499A, we moved from reading existing methods to implementing four research baselines: SecureDyn-FL, FedMADE, Weight Evolving Frequency (WEF) defense, and FedMSE. Each baseline addresses a different weakness in federated IDS design—privacy and poisoning, class imbalance, free-rider participation, or limited attack labels. Running these methods gave us a clearer view of where they agree, where their assumptions differ, and which parts can realistically support the next stage of the thesis. One issue stands out: a useful federated IDS must separate unusual but legitimate non-IID clients from harmful participants without discarding minority-attack knowledge or exposing private client information. CSE 499B will therefore begin with controlled, common experiments and use those results to study how class-aware and trust-aware evidence can be combined without turning the system into an untested collection of defenses.

I. INTRODUCTION AND REVIEW PURPOSE

IoT networks rarely behave like a single, uniform data source. Cameras, sensors, gateways, and smart appliances produce different traffic patterns and face different attacks. A centralized intrusion-detection system (IDS) can collect that traffic in one place, but doing so exposes sensitive telemetry, adds communication cost, and ties detection to a central service. Federated learning (FL) avoids routine raw-data transfer by training locally and sharing model updates. That shift solves one problem but introduces others. Updates can still leak information, non-IID clients can pull training in different directions, compromised participants can poison the global model, and some clients may benefit from the federation without performing meaningful local training [1]–[6].

PRIVATE SHIELD began as an attempt to study these problems within one experimental system. During CSE 499A, we first built a basic federated pipeline, then implemented baselines drawn from SecureDyn-FL, FedMADE, WEF-Defense, and FedMSE. Working with the code changed the purpose of the literature review. The papers were no longer only references; they became test cases that exposed different ways of judging a client contribution. This CSE 499B review records what those implementations taught us, where the assumptions conflict, and which unresolved problem is worth carrying forward as thesis work.

II. PROBLEM STATEMENT

A practical federated IDS has to make several decisions at once. It must learn from clients whose traffic is genuinely different, protect local information, retain knowledge of rare attacks, and limit the influence of clients that are malicious, unstable, or simply not contributing. Existing methods usually solve one or two of these issues well, but the signals they use can disagree. A benign device with unusual non-IID traffic may look suspicious to a poisoning detector. A malicious or free-riding client, by contrast, may return an update that stays close to the normal population and escapes a simple distance check. Standard averaging creates another problem: a client that carries useful knowledge about a rare attack may receive little influence because it owns relatively few samples.

For this thesis, high global accuracy is therefore not enough. The harder question is which client knowledge the server should trust, how much influence it should receive, and whether that

influence should depend on the attack class being learned. Any proposed method must be judged with per-class results, malicious-client rejection, benign false rejection, convergence behavior, privacy cost, and communication overhead—not with one headline accuracy value.

III. REVIEW OF IMPLEMENTED RESEARCH METHODS

A. SecureDyn-FL: Privacy, Poisoning, and Heterogeneity

SecureDyn-FL provides the broadest security-oriented reference among the methods we studied. It combines temporal gradient auditing, privacy-preserving aggregation, personalization for non-IID clients, and pruning and quantization to reduce communication cost [1]. Its poisoning detector tracks client updates over time with a Gaussian Mixture Model (GMM) and Mahalanobis distance. The framework also separates a shared feature extractor from a personalized classifier, allowing local clients to adapt to different class distributions while still contributing to global training.

We used SecureDyn-FL as the starting point for PRIVATE SHIELD’s federated pipeline. The CSE 499A implementation covered N-BaIoT-compatible preprocessing, simulated client partitions, local 1D-CNN training, server-side averaging, and basic IID/non-IID runs. The full mechanism from the source paper was not reproduced. GMM/Mahalanobis auditing, encrypted aggregation, personalization, pruning, quantization, and complete poisoning experiments remain later targets. Keeping that boundary explicit matters: our dataset portions, hardware, and implementation choices differ from the conditions reported in the paper.

B. FedMADE: Class-Aware Dynamic Aggregation

FedMADE addresses a problem that became especially relevant once we began looking beyond global accuracy: device-specific class imbalance. Instead of giving clients influence mainly according to sample count, FedMADE evaluates each local model with a Class Probability Matrix (CPM). It then groups similar class-level behavior with DBSCAN and computes aggregation weights from the contribution of each cluster [2]. This matters when an attack appears on only a few devices. Under ordinary averaging, knowledge from those clients can be diluted even though it may be the only useful signal for that class.

Our FedMADE baseline implements the main steps needed for comparison: local-model evaluation, CPM construction, DBSCAN grouping, dynamic weight calculation, and down-weighting of suspicious models. In practice, this gave us a second view of a client that SecureDyn-FL does not provide. Two clients can look similar in parameter space yet behave very differently on a specific attack class. The method also raised a new question for the next semester: CPMs depend on a small server-side auxiliary validation set, so the quality and privacy of that reference data can directly affect the aggregation decision.

C. WEF-Defense: Detecting Free Riders

Free riding is different from direct poisoning. A selfish client can receive the global model while doing little meaningful local training, and its returned parameters may not look extreme enough for a standard poisoning detector. WEF approaches

the problem from the training process itself [3]. It tracks how frequently selected weights change during local optimization. Genuine training should leave a recognizable pattern of weight evolution; copying or minimally modifying a model should produce a different one. Distance and similarity measures over WEF matrices can then separate ordinary contributors from suspected free riders.

Our implementation tracks changes in penultimate-layer weights, constructs WEF matrices, and compares clients using Euclidean distance, cosine similarity, and average evolving frequency. The experiment made one point clear: update magnitude by itself does not tell us whether useful work occurred on the client. WEF is still only evidence, however. A benign device with limited resources or unstable training can also produce an unusual evolution pattern, so the server should not treat one score as final proof of misconduct.

D. FedMSE: Semi-Supervised Anomaly Detection

FedMSE gave the project a different detection model altogether. Rather than requiring labeled examples of every attack, it learns normal traffic with a Shrink Autoencoder and a centroid-based one-class classifier [4]. During aggregation, MSEAvg favors local models with lower reconstruction error. This setup is useful when attack labels are incomplete or when the test set contains behavior that was not represented during training.

The PRIVATE SHIELD baseline follows the core FedMSE workflow with N-BaIoT-compatible data, non-IID gateway partitions, normal-only training, centroid-distance anomaly scoring, and MSEAvg-style weighting. Comparing it with FedMADE was useful because the two methods ask different questions. FedMADE measures how well a client handles known classes; FedMSE measures how consistently a client represents normal behavior. That contrast helped us separate known-class discrimination from sensitivity to previously unseen anomalies.

IV. IMPLEMENTATION STATUS AND LESSONS FROM CSE 499A

A. SecureDyn-FL implementation status

By the end of CSE 499A, the basic SecureDyn-FL-oriented federated reference was running end to end. We verified preprocessing, client-wise partitioning, simulated local training, server aggregation, and IID/non-IID execution. We did not claim the source paper’s advanced auditing, cryptographic aggregation, personalization, pruning, quantization, or complete poisoning results. Those components remain implementation targets for controlled experiments in CSE 499B.

B. FedMADE implementation status

The FedMADE baseline is also operational at the level needed for research comparison. The server forms CPMs from local models, groups similar class-level behavior with DBSCAN, calculates dynamic aggregation coefficients, and can reduce the weight of suspicious models. The most useful lesson came from a simple observation: dataset size and client usefulness are not the same thing. A small client may carry the only strong representation of a rare attack and can lose that contribution under FedAvg.

C. WEF-Defense and FedMSE implementation status

WEF-Defense now gives us a training-behavior signal based on weight evolution, while FedMSE gives us a semi-supervised signal based on normal-only representation learning, reconstruction error, and centroid distance. Both implementations capture the core research idea rather than every configuration used in the source papers. Together, they pushed PRIVATE

SHIELD beyond a design that relies only on parameter distance to judge clients.

Running the four baselines side by side changed how we framed the project. They are not modules that should automatically be stacked into one system. Each one answers a different question. FedAvg mostly rewards data volume. FedMADE asks whether a client contributes useful class knowledge. WEF asks whether meaningful local optimization appears to have happened. FedMSE asks whether normal traffic is modeled consistently. SecureDyn-FL adds temporal evidence about suspicious updates. Any combined design should use only signals that add distinct information under the same experimental conditions.

The second lesson was methodological. The source papers use different datasets, client counts, partitions, architectures, and attack assumptions, so their reported numbers are not directly comparable. In CSE 499B, we need common data splits, random seeds, participation rules, local epochs, communication rounds, and attack schedules wherever the methods allow it. Published results will remain reference points; experimental claims in our report will come only from our own runs.

V. COMPARATIVE ANALYSIS OF THE FOUR METHODS

A. What Each Method Treats as Evidence

The clearest difference among the four methods is the evidence each one trusts. SecureDyn-FL looks at the temporal and geometric behavior of model updates: an update becomes suspicious when its trajectory departs from the evolving benign population. FedMADE looks at class behavior instead; its CPM acts as a semantic performance signature rather than a parameter-distance signature. WEF checks whether local optimization appears to have occurred, while FedMSE checks how well a client models normal traffic and detects deviation. These signals can disagree. A client may be unusual in parameter space yet valuable for a rare class, or look ordinary in parameter space while quietly harming that class.

That disagreement is not an edge case in IoT networks; it is expected. Different device types naturally produce different feature distributions, attack frequencies, and resource constraints. A server cannot safely assume that the majority update is always the right one. Our implementation work therefore points toward evidence-based aggregation, where each signal is tested against the failure mode it is meant to detect rather than treated as a universal trust score.

B. Assumptions About Data and the Server

SecureDyn-FL assumes that client data stay local and that even the server should learn as little as possible about individual updates. Its design therefore emphasizes encrypted aggregation and privacy-aware auditing. FedMADE makes a different assumption: the server evaluates local models using a small auxiliary set that contains examples from the attack classes. That allows class-aware weighting, but it also makes the decision dependent on how representative those auxiliary samples are, especially for rare attacks. WEF requires training-behavior evidence from model-weight changes, while FedMSE assumes that normal-only learning provides a meaningful basis for anomaly detection.

These assumptions become visible as soon as we try to combine the methods. A privacy design that hides individual models from the server can conflict with a class-aware method that expects direct model evaluation. A legitimate distribution shift can also make a client look suspicious to a temporal auditor even when the new data are useful. Before we combine any modules in CSE 499B, we therefore need to define exactly what the server can observe and why each piece of metadata is necessary.

C. Robustness Coverage

The methods also cover different failure modes. SecureDyn-FL focuses on poisoning and privacy threats. FedMADE is aimed mainly at heterogeneous class imbalance, although poor-performing models can receive less influence. WEF targets free riders whose updates may remain close to ordinary clients, and FedMSE addresses limited attack labels by learning normal traffic. None of these methods alone resolves adaptive collusion, noisy trust evidence, minority-targeted stealth attacks, or the risk of penalizing a benign non-IID client.

That gap changes the thesis question. Asking which algorithm is ‘best’ is less useful than asking which evidence remains dependable under a specific threat. A meaningful comparison should show whether malicious clients were actually identified, how often benign clients were rejected, whether rare-class recall survived the attack, and how performance changed as the malicious-client ratio increased. Global accuracy belongs in that analysis, but it cannot carry the argument by itself.

D. Efficiency and Deployability

Cost matters because the intended setting is IoT. SecureDyn-FL includes pruning and quantization partly because encryption and auditing add computation and communication overhead. FedMADE keeps its clustering and weight calculation on the server, which limits extra work on constrained clients, but the server still has to evaluate local models repeatedly. WEF adds weight monitoring during local epochs, and FedMSE adds autoencoder-based representation learning. These choices are reasonable for a research prototype only if we measure what they cost.

CSE 499B experiments should therefore record round duration, communication volume where available, memory or execution observations, and the extra cost introduced by each defense. A method that preserves a rare class but sharply increases communication or falsely rejects many benign clients may not be practical for an IoT deployment. Efficiency is part of the research question, not something to discuss only after the model results look good.

VI. RESEARCH GAPS IDENTIFIED FROM THE REVIEW

A. Benign Heterogeneity versus Malicious Behavior

An unusual client update is not automatically an attack. Cameras, sensors, gateways, and smart appliances can produce very different traffic even when every device is honest. A detector that relies too heavily on parameter distance may reject a rare but useful non-IID client. The same client may also carry minority-class knowledge that FedMADE would want the global model to keep.

B. Global Accuracy Can Hide Minority-Class Failure

Global accuracy can look healthy while a rare attack class quietly collapses. That creates a real blind spot: an attacker may target one minority class without causing an obvious drop in the overall score. FedMADE can expose class-level damage, while SecureDyn-FL can expose suspicious temporal update behavior. Reading and implementing both methods suggests a more focused question—can the server judge a client by both how its update behaves and which class knowledge it changes?

C. Trust Signals Can Be Noisy

None of the available trust signals is perfectly clean. WEF can react to benign instability, FedMSE can flag unusual but legitimate traffic, and FedMADE-style auxiliary data may be incomplete or shifted. A permanent binary label such as ‘trusted’ or ‘malicious’ is therefore too rigid for the cases we want to study. Round-level scores, explicit false-rejection

reporting, and ablation experiments offer a better way to show which evidence actually drives each decision.

VII. PRELIMINARY CSE 499B RESEARCH DIRECTION

Based on the CSE 499A implementations, our preliminary direction is a trust-aware, class-aware aggregation method that combines SecureDyn-FL-style temporal security evidence with FedMADE-style class contribution evidence. We are especially interested in a stealthy attack in which one or more compromised clients keep their updates close to benign behavior while degrading a minority attack class. The global accuracy may still look acceptable, which makes the attack easy to miss if the server only watches aggregate metrics.

We will not begin by merging every mechanism at once. The first step is to run FedAvg, the SecureDyn-oriented baseline, and FedMADE under a common setup, then introduce fixed malicious-client ratios and measure both update-space behavior and per-class performance. The main research question is whether a multi-view client score can protect minority-class performance without rejecting legitimate non-IID clients. If that direction proves unstable or too complex, the same review leaves two defensible alternatives: privacy-preserving class-aware aggregation and class-aware aggregation that remains reliable when the auxiliary reference data are incomplete or untrustworthy.

VIII. DATASETS AND EVALUATION PLAN

The evaluation plan is designed to answer the research question rather than to maximize a single score. We will track per-class behavior alongside client-level security decisions and system cost. Accuracy will remain a supporting metric, but macro-F1, class-wise precision/recall/F1, malicious-client detection or rejection, benign false rejection, convergence, and model degradation under attack will carry more weight. Where measurable, we will also record round time, communication cost, and execution or memory observations.

Ablation studies will separate the effect of each signal. The comparison will include sample-count aggregation alone, class-aware aggregation alone, temporal/adversarial trust alone, and the combined mechanism. Fixed random seeds and repeated runs will help distinguish genuine defense effects from normal training variation.

A. Primary and Secondary Datasets

CICIoT2023 is the preferred primary dataset for the class-aware experiments because its attack categories are unevenly distributed across a large IoT environment. That matches the setting FedMADE was designed to address: minority attacks can disappear during aggregation when influence follows sample count. N-BaIoT provides a useful second dataset because it contains device-level benign and botnet traffic and already fits the SecureDyn-oriented and FedMSE-oriented preprocessing explored in CSE 499A. Testing on both datasets will also show whether a trust rule is overly tied to one traffic distribution.

B. Experimental Scenarios

We will start with clean IID and non-IID runs to establish the normal range of client variation. Controlled poisoning will then introduce label flipping, scaled or randomized updates, and model replacement at fixed malicious-client ratios. Free-rider experiments will remain separate because selfish behavior is not the same as an adversarial update. For the class-aware thesis direction, the main stress test will be a stealthy attack that selectively harms a rare class while trying to keep global accuracy and overall update statistics within an apparently normal range.

C. Metrics and Reproducibility

Each experiment will report accuracy, precision, recall, macro-F1, per-class F1, and AUC where the detector supports it. Security results will include malicious-client detection or rejection, benign false rejection, poisoning-induced model degradation or attack success, and recovery across communication rounds. If differential privacy is enabled, epsilon and delta will be reported with utility metrics. To make the runs reproducible, we will record the dataset portion, feature set, preprocessing order, client partition rule, random seed, local epochs, batch size, participation rate, number of rounds, and every active attack or defense.

The key comparison will be the ablation study. FedAvg will provide the sample-count baseline, FedMADE will isolate the effect of class-aware aggregation, and the SecureDyn-oriented audit will isolate temporal/adversarial validation. We will then remove one component at a time from the combined method. That design should show whether any gain comes from the interaction between trust and class utility rather than from adding complexity to the pipeline.

IX. CONCLUSION

CSE 499A moved PRIVATE SHIELD beyond the proposal stage and gave us four working research baselines with different views of federated IoT intrusion detection. SecureDyn-FL frames the problem around privacy, poisoning, and heterogeneous clients. FedMADE shows why class-aware aggregation matters when rare attacks appear on only a few devices. WEF adds evidence about whether a client actually trained, and

FedMSE adds a normal-only anomaly-detection path. The next step is not to add another paper to the stack. We need a common experimental setup, careful comparison, and a focused study of how trust evidence and class utility should interact. That progression gives CSE 499B a clear research question grounded in what we have already built and tested.

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